

Nicholas Anderson

Senior Data Engineer at CME Group | U.S. Citizen

nanderson.works@outlook.com | (872) 261-2089 | Cape Coral, FL

PROFESSIONAL SUMMARY

Senior Data Engineer with **13+ years** of experience architecting large-scale **ETL/ELT**, real-time streaming, and **lakehouse** platforms across **fintech**, **healthcare**, and **SaaS ecosystems**. Expert in building cloud-native data systems using **AWS**, **Azure**, **Apache Spark**, **Kafka**, **Airflow**, **dbt**, **Delta Lake**, **Iceberg**, and modern DevOps tooling. Proven leader in **driving modernization**, **governance**, and **observability** through **Terraform**, **Kubernetes**, **Datadog**, **Prometheus**, **Great Expectations**, and **Apache Atlas**. Recognized for delivering high-reliability pipelines, optimizing compute/storage efficiency, and partnering with quant, risk, and analytics teams to ship mission-critical, high-fidelity data products.

TECHNICAL SKILLS

- **Languages:** Python, Bash, PowerShell, Go, Java
- **Data Processing, ETL & Orchestration:** Apache Spark, Apache Beam, Apache Airflow, dbt
- **Cloud Data Platforms:** AWS (S3, Glue, Lambda, Athena, Redshift, Kinesis), Azure (Data Factory, Synapse Analytics, Blob Storage, Event Hubs)
- **Data Lakes & Storage:** Delta Lake, Apache Iceberg
- **Streaming & Real-Time Systems:** Apache Kafka, Apache Flink
- **Data Warehousing:** Snowflake, Amazon Redshift
- **DevOps, Infrastructure & CI/CD:** Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps, GitLab CI
- **Monitoring & Observability:** Prometheus, Grafana, Datadog, ELK Stack (Elasticsearch)
- **Data Governance, Quality:** Great Expectations, Apache Atlas, Data Contracts & schema governance
- **Analytics & BI:** Tableau, Looker, Power BI

WORK EXPERIENCE

CME Group | Chicago, IL

Aug 2022 – Present

Senior Data Engineer

- Led end-to-end delivery of a completed **fintech market-data** and **risk-analytics platform**, architecting scalable pipelines using **Python**, **Apache Spark**, and **AWS** services.
- Architected real-time streaming solutions using **Apache Kafka** and **Kinesis**, supporting sub-second event propagation for trading, surveillance, and liquidity-risk systems.
- Designed cloud-native ingestion frameworks on **AWS S3**, **Glue**, **Lambda**, **Athena**, and **Redshift**, enabling high-volume processing of trade, quote, and reference data.
- Established **lakehouse patterns** with **Delta Lake** and **Apache Iceberg**, improving schema evolution, ACID guarantees, and long-term data reliability.
- Directed orchestration strategy using **Apache Airflow**, implementing SLA-driven DAGs, automated retries, and dependency-aware scheduling for mission-critical fintech workloads.

- Built enterprise-grade transformation layers with **dbt**, standardizing metrics, lineage, and documentation across quant, risk, and clearing teams.
- Drove infrastructure modernization using **Terraform**, **Docker**, and **Kubernetes**, enabling reproducible deployments and reducing environment drift across clusters.
- Implemented advanced observability with Prometheus, **Grafana**, and **Datadog**, creating latency dashboards, anomaly alerts, and pipeline-health SLOs.
- Configured **Datadog APM** for **Spark**, **Airflow**, and **microservices**, improving root-cause analysis and reducing MTTR across critical data systems.
- Built collaborative analytics environments using **Databricks Repos**, **MLflow**, and **Unity Catalog**, standardizing governance and access controls across teams.
- Strengthened **data governance** with **Great Expectations**, **Apache Atlas**, and **data-contract frameworks**, ensuring regulatory alignment for audit, compliance, and market-integrity requirements.
- Collaborated with quantitative researchers and risk engineers to deliver high-fidelity datasets for pricing models, volatility surfaces, and market-microstructure analytics.
- Optimized compute, storage, and partitioning strategies across **AWS** workloads, reducing operational cost and improving pipeline throughput by 45%.
- Mentored junior engineers, established coding standards, and led design reviews for scalable, secure, and compliant data-engineering solutions.

TEKsystems | Orlando, FL

Oct 2017 – Jul 2022

Data Engineer

- Engineered large-scale **ETL/ELT pipelines** on **Azure Cloud** using **Apache Spark**, **Airflow**, and **Python**, processing millions of **healthcare** and **e-commerce** records daily while ensuring high availability, scalability, and data integrity.
- Designed and deployed cloud-native data services on **Azure (Data Factory, Synapse, Event Hubs)** to support cross-platform analytics.
- Built real-time streaming pipelines with **Apache Kafka** and **Flink**, enabling sub-second event ingestion for **e-commerce** order and inventory systems.
- Designed and optimized large-scale **ETL/ELT pipelines** on **Databricks** using **PySpark**, **Delta Lake**, and **SQL**, improving processing throughput and reliability for high-volume datasets.
- Modernized legacy **ETL** jobs by migrating to **Airflow 2.1**, reducing pipeline failures by **45%** and improving SLA reliability.
- Developed secure data ingestion frameworks using **Azure Data Factory**, orchestrating pipelines across **Blob Storage**, **Event Hubs**, and **Synapse Analytics**.
- Migrated legacy ETL workloads to **Snowflake** and **Azure Synapse**, reducing compute cost and improving reliability across analytics teams.
- Developed analytical datasets in Snowflake to support BI tools such as **Power BI**, **Tableau**, and **Looker**.
- Integrated **Snowflake** with **Azure Event Hubs** and **Kafka** for near-real-time ingestion of **healthcare** event stream
- Implemented **Delta Lake** and **Apache Iceberg**–based lakehouse patterns to unify batch and streaming workloads across healthcare datasets.
- Developed scalable APIs and microservices using **Go** and **Python**, supporting secure data exchange across regulated environments.
- Optimized **SQL** and **Spark** jobs, reducing compute cost by **30%** and improving end-to-end pipeline latency.
- Migrated legacy reporting pipelines to **dbt**, standardizing transformations and improving developer productivity.

- Delivered **HIPAA**-aligned data ingestion frameworks with **Great Expectations** for automated validation and quality enforcement.
- Built CI/CD workflows using **GitHub Actions**, and **GitLab CI**, automating testing, packaging, and infrastructure deployment.
- Containerized data services using **Docker** and deployed them on **Kubernetes**, improving environment consistency and deployment speed.
- Implemented metadata and lineage tracking with **Apache Atlas**, enabling full governance across **healthcare** and **e-commerce** domains.
- Upgraded distributed processing workloads to **Spark**, unlocking adaptive query execution and improving performance on large joins.
- Partnered with product, analytics, and engineering teams to deliver end-to-end data solutions powering both healthcare insights and **e-commerce** personalization engines.

AgileEngine | Boca Raton, FL

Jul 2013 – Sept 2017

Data Analystist

- Analyzed large, multi-source datasets within a **SaaS platform** using **SQL**, **Python**, and **Excel** to identify trends, anomalies, and actionable insights that informed product strategy, optimized user engagement, and improved operational efficiency.
- Designed and maintained dynamic dashboards and reports using **Tableau**, **Power BI**, and **MySQL 5.7**, improving visibility into customer behavior, churn, and product usage metrics across departments.
- Built automated **ETL pipelines** leveraging **Python 3.6**, **Pandas**, **REST APIs**, and **PostgreSQL 9.6** to extract, clean, and transform SaaS application logs and transactional data for analytics and reporting.
- Partnered with engineering teams to define structured data requirements and implemented consistent tracking using **SQL**, **JSON**, and **API integrations**, ensuring accurate and scalable data capture across the **SaaS platform**.
- Developed KPI frameworks and standardized metric definitions using **SQL**, **Excel**, and **Power BI**, aligning reporting across product, finance, and customer success teams.
- Conducted cohort, retention, and funnel analysis with **SQL**, **Python**, and statistical modeling techniques to support roadmap prioritization and feature optimization decisions.
- Improved data quality by implementing validation rules, reconciliation checks, and structured data dictionaries within **PostgreSQL** and **ETL pipelines**, increasing reporting accuracy and stakeholder trust.
- Translated complex business requirements into analytical models and executive-level insights using **Tableau**, **Power BI**, and advanced **SQL queries**, accelerating data-driven decision-making.
- Performed A/B test analysis and hypothesis testing using **Python (SciPy, NumPy)** and **SQL**, quantifying feature impact and guiding product optimization strategies.
- Delivered analytics requirements, usage insights, and performance reporting that directly supported the successful launch and adoption growth of the **SaaS platform**.
- Documented end-to-end data workflows, transformation logic, and reporting methodologies to ensure maintainability, auditability, and knowledge transfer across technical teams.

EDUCATION

University of Florida | Gainesville, FL

Sept 2009 – Jun 2013

Bachelor's degree in Computer Science